孫正義育英財団

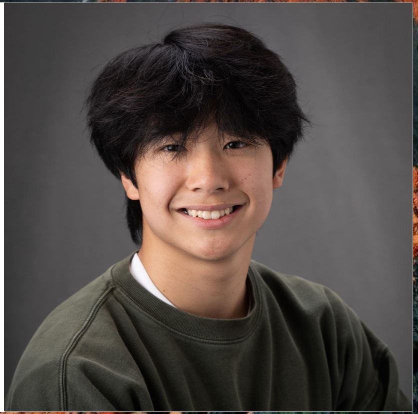
Generation 9

ADAM CHEONG

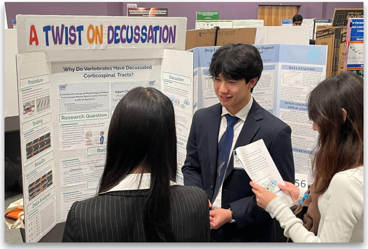
BORN IN 2009

READ MORE

#robotics
#engineering
#mathematics



Masason Foundation



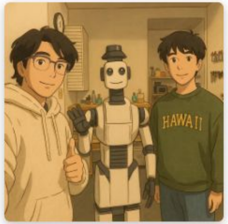
Adam Cheong

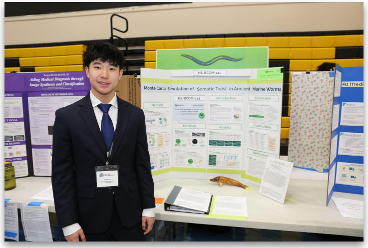
Proof School | San Francisco, California | adam@cheong.net

A high school sophomore from the San Francisco Bay Area with a strong interest in science, engineering, math and finance, Adam interns afterschool at a robotics startup in San Francisco. Adam aspires to identify and formulate the most important unsolved problems in robotics for his generation, and to create a future in which collective intelligence and resources are mobilized to solve them.

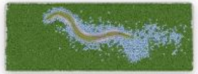
Adam practices bouldering, a unique skill that combines puzzle solving with physical feats — exercising both brain and brawn — as the climber places one foothold after another to chart a navigable course. From mechanical engineering to computational biology, Adam's projects since 7th grade have won STEM fair awards every year.

Adam is taking a computer science class this summer at UC Berkeley as a pre-college scholar, joining his older brother there. Adam plans to earn a pilot license before heading out to college, where he will double major in engineering and finance.





Accepted Abstract



Monte Carlo Simulation of Somatic Twist in Ancient Marine Worms

Adam Cheong

25th International Worm Meeting, UC Davis, Jun 2025

ABS PDF

Monte Carlo simulation to recreate evolutionary events inside a computer in order to observe how the first digital worm organism evolves to become the first fish when given bountiful food sources.

Submitted Paper

Why Do Vertebrates Have Decussated Corticospinal Tracts?

Adam Cheong

Junior Science & Humanities Symposium, San Francisco State University, Feb 2025

ABS PDF

A comparative study of morphological models from ancient marine worms to the first fish inside an aquarium.

Submitted Essay

The Rise of Homo Roboticus

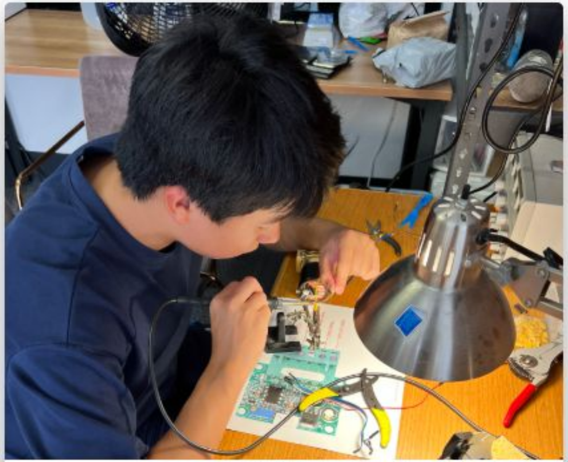
Adam Cheong

Masason Foundation, Jan 2025

ABS PDF

10 years later, how will artificial intelligence be further developed and what kind of changes will it bring to the world?

work



watney robotics

autonomous physical infrastructure

💪 fold,fold,fold!

dynamic duo working night shifts

🍓 hydroponics

set up vertical farming @ Berkeley

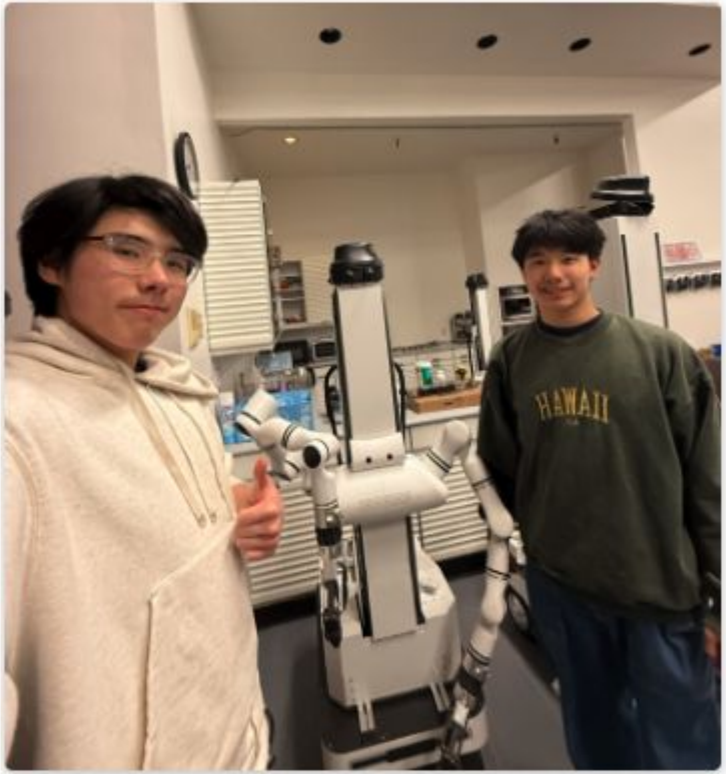
watney robotics

autonomous physical infrastructure

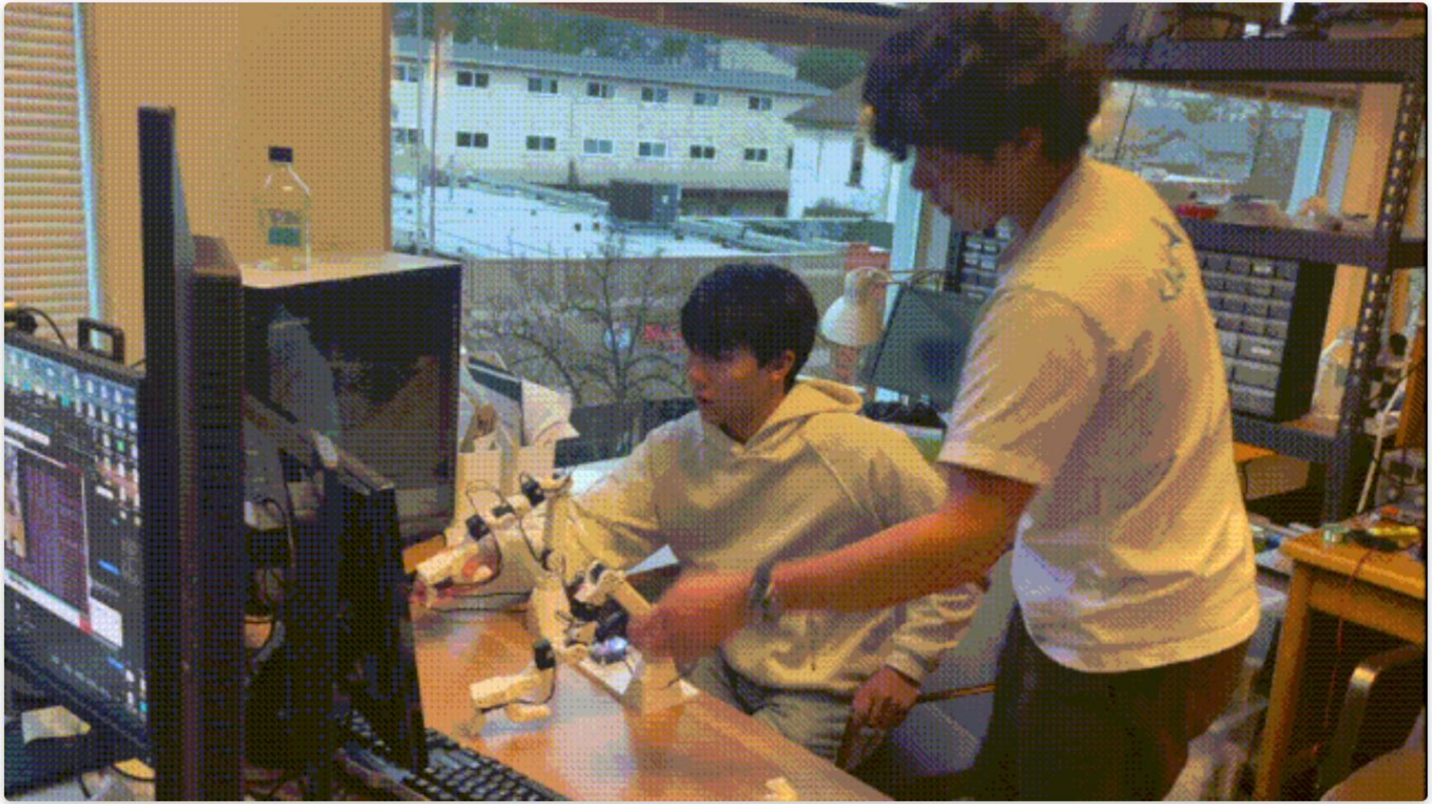
My brother’s big adventure in embodied AI — so glad we could be on this quest together!

Watney alive: goes to work, cleans window, and goes shopping (circa July–Aug 2024).

Watney making itself useful poolside and in the laundry room folding towels (circa Aug–Nov 2024).



(left) Meet Watney's creator my eldest brother (circa April 2025); (right) our new offices in the SF Mission District (circa Jun 2024).



【 LINK 🗣️ 】 Closing an ultra low-latency teleoperation loop at our Berkeley offices on Telegraph Avenue (circa Feb 14, 2024).

🦾 Watney continuously learned to become a better 🍓 hydroponics farmer (circa Nov–Dec 2023).

Our zero-to-one journey: starting 🍓 hydroponics in Berkeley (circa Aug–Oct 2023).

Heavy lifting all the way: from San Carlos home garage to new offices in Berkeley (circa July 2023).

"You can't connect the dots looking forward; you can only connect them looking backwards. So you have to trust that the dots will somehow connect in your future." — Steve Jobs (2005 Stanford commencement address).

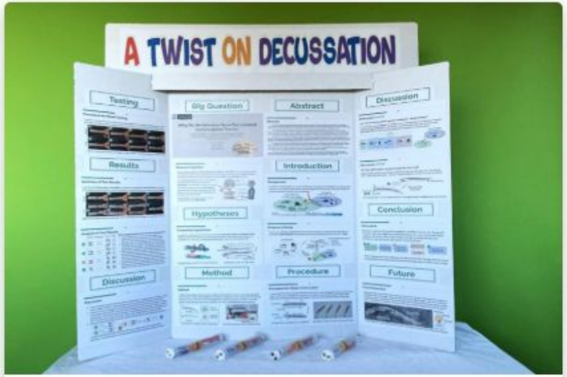
Adam Cheong

about | publications | **projects** | cv

explore

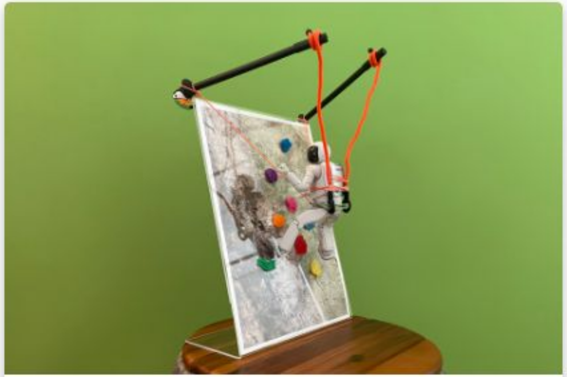
🏅 monte carlo

Monte Carlo simulation of somatic twist in ancient marine worms



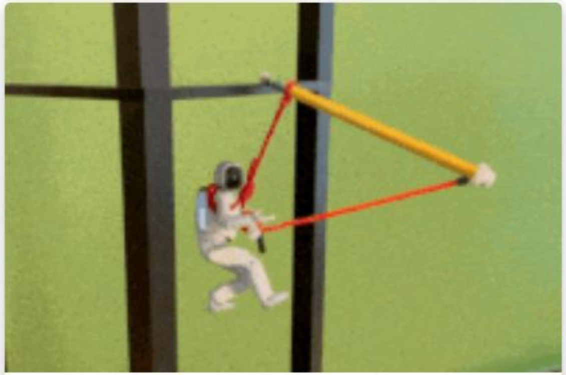
🥇 decussation

why do vertebrates have decussated cortocospinal tracts?



🥇 cliffhanger v2

dynamically stable mechanical mountain climbing aid by design fun



🧗 cliffhanger

dynamically stable mechanical mountain climbing aid by design

🚀 rocket fuel

homemade rocket fuel with kitchen chemistry (i.e., kno3 + sugar etc.)

🏎️ go kart go!

General
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CV

PDF

General

Name	Adam Cheong
Goal	To identify and formulate the most important unsolved problems in robotics for our generation, and to create a future in which collective intelligence and resources are mobilized to solve them.
Email	adam@cheong.net
Phone	(650) 593-8322
Url	https://adamcheong.com
Summary	Inventive high school student from San Francisco with a strong interest in science, engineering, math and finance; winner of annual STEM fair awards; natural born athlete who loves bouldering and plays the electric guitar.

Education

2023.08 - 2027.06
San Francisco, California

High School Diploma

Proof School

9th to 12th Grade

- 10th: Literary Arts: Graphic Narrative
- 10th: AP Calculus BC
- 10th: AP Physics C: Mechanics
- 10th: AP Chemistry
- 10th: Latin 2
- 9th: Latin 1
- 9th: AP Biology
- 9th: Advanced Literature
- 9th: Exponents/Log/Trig A & B
- 9th: Euclidean Geometry 3
- 9th: Number Theory 2
- 9th: Graph Theory 1
- 9th: Studio Art: Paper

4 Years of Advanced Math Curriculum

Year	Block 1	Block 2	Block 3	Block 4	Block 5
9th	Graph Theory	Exp/Logs/Trig A	Geometry 3	Exp/Logs/Trig B	Number Theory 2
10th	Discrete Probability	Differential Calculus	Integral Calculus	Series and Topics in Calculus	Number Theory 3
11th	Combina-torics 2*	Linear Algebra A*	Linear Algebra B*	Mathematical Logic*	Cryptography*
12th	Theory of Partitions*	Group Theory*	Statistics A / Multivariable Calculus A*	Statistics B / Multivariable Calculus B*	Special Relativity*

2017.08 - 2023.06
Portola Valley, California

Middle School

Woodland School

3rd to 8th Grade

- Capstone Project: Hydroponics
- Mathcounts, AMC 8, and STEM Fairs

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Experience

2023.07 - 2027.06

San Francisco, California

Robotics Intern

Watney Robotics

Assemble robot arms and controllers that allow humans from halfway around the world to teleoperate a robot; perform data labeling and ML models training.

- Robot Arm Controller Assembly
- Teleoperation Testing
- Data Labeling
- ML Models Training

Volunteering

2022.07 - 2022.07

Portola Valley, California

Camp Counselor

Woodland Summer Adventures

Supervised students and gained first hand experience teaching small groups of students a new skill.

- Leadership Development
- Safety Training
- CPR, AED & Basic First Aid Certified

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Projects

2025.01 - 2025.12

Monte Carlo Simulation of Somatic Twist in Ancient Marine Worms

Research Purpose: run Monte Carlo simulation to recreate evolutionary events inside a computer and see how the first digital worm evolves in silico to become the 'first fish' with a dorsoventrally inverted body plan after a somatic twist.

- Computational Biology
- OpenWorm
- Sybernetic

2023.01 - 2023.03

The Cliffhanger Project

A novel mountain climbing aid that brings a new mechanical advantage to mountaineers by means of a simple lever mechanism — inspired by a physics demo harnessing a setup of three toothpicks to carry a payload hanging off the edge of a table.

- Mountain Climbing Aid
- Mechanical Advantage
- Type 2 Lever

- General
- Education
- Experience
- Volunteering
- Projects
- **Science fairs**
- Contests
- Posters
- Papers
- Courses
- Societies & clubs
- Skills
- Languages
- Assessments

Science fairs

2025.03

4th Place, Biostatistics and Computational Biology

High School Division, Alameda County Science and Engineering Fair (ACSEF)

Monte Carlo Simulation of Somatic Twist in Ancient Marine Worms

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2024.04

Honorable Mention, Zoology

Senior Division, California Science and Engineering Fair (CSEF)

Why Do Vertebrates Have Decussated Corticospinal Tracts?

2024.03

1st Place & Bill Tobin Award, Life Sciences

9th Grade, Golden Gate STEM Fair (GGSF)

Why Do Vertebrates Have Decussated Corticospinal Tracts?

2023.04

Honorable Mention, Applied Mechanics and Structures

Junior Division, California Science and Engineering Fair (CSEF)

Cliffhanger: A Dynamically Stable Mechanical Mountain Climbing Aid by Design

2023.03

1st Place, Engineering: Electrical and Mechanical

8th Grade, Golden Gate STEM Fair (GGSF)

Cliffhanger: Dynamic Stability in a Mechanical Mountain Climbing Aid by Design

2023.03

1st Place, Engineering, Technology and Application of Science

8th Grade, San Mateo STEM Fair

Cliffhanger: Dynamic Stability in a Mechanical Mountain Climbing Aid by Design

2022.03

Honorable Mention, Biological Systems

7th Grade, San Mateo STEM Fair

Does the Color of Light Affect Plant Growth?

Contests

2025.03

Honorable Mention

USA Biolympiad (USABO)

The most prestigious biology education and testing program for US high school students, participated by approximately 10,000 talented students annually.

2024.04

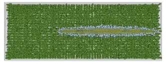
Maxima Cum Laude (Silver Recognition)

National Latin Examination

Annual recognition of student accomplishment in Latin language learning and the study of Greco-Roman culture.

Posters

2025.06.28

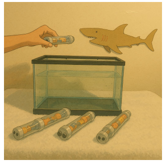


Monte Carlo Simulation of Somatic Twist in Ancient Marine Worms

25th International Worm Meeting, UC Davis

Progress report on design aspects of evolutionary modeling for a digital worm and computational challenges encountered.

2025.02.28



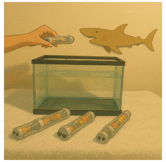
A Twist on Decussation

Junior Science and Humanities Symposium, San Francisco State University

A comparative study of morphological models from ancient marine worms to the first fish inside an aquarium.

Papers

2025.02.28



Why Do Vertebrates Have Decussated Corticospinal Tracts?

Junior Science and Humanities Symposium, San Francisco State University

A comparative study of morphological models from ancient marine worms to the first fish inside an aquarium.

2025.01.25

The Rise of Homo Roboticus

Thesis Submission, Masason Foundation

10 years later, how will artificial intelligence be further developed and what kind of changes will it bring to the world?

2023.06.30

People Get Essays Wrong

Global Essay Competition, John Locke Institute

What is something important, about which nearly everybody is wrong?

Courses

The Structure and Interpretation of Computer Programs
UC Berkeley Pre-College Scholar 2025-06

The Mathematics of Competitive Behavior
Johns Hopkins Center of Talented Youth 2024-08

Engineering and Robotics Workshop at MIT Campus
National Geographic Student Travel 2024-07

The Concord Review History Camp
The Concord Review 2024-06

Logic: Principles of Reasoning
Johns Hopkins Center of Talented Youth 2023-07

Societies & clubs

Genetics Society of America

(2025—present) K-12 member of an international community of 6,000 biologists from 50 countries advancing the field of genetics.

Berkeley Math Circle

(2015–present) Intermediate level II member; attend evening math lectures by professors at UC Berkeley campus every Wednesday.

Benchmark Climbing

(2023–present) Member of afterschool climbing club; practice 2 to 3 times a week at an indoor bouldering gym in San Francisco.

Clock Tower Music

(2023–present) Taking guitar lessons with a local musician afterschool; practice new songs in the evenings.

24 Hour Fitness

(2024–present) Member of a neighborhood gym where I workout regularly in the evenings and on weekends.

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Skills

Hardware

Robot Arm Controller Assembly
Circuit Board Soldering
PC Assembly

Software

Python
Data Labeling
ML Models Training

DIY

Hydroponics
3D Printing
Origami

Investment

Trading Strategies
Trading Tools
Risk Management

Languages

Latin
learning (2 years)

Japanese
limited phrases

Mandarin
limited phrases

Spanish
conversational (6 years)

English
native speaker fluency

Assessments

SAT (10th Grade)
Score: 1510 (730 EBRW, 780 Math)

SCAT (8th Grade)
CTY High Honors (484 Verbal, 501 Math)

STB (8th Grade)
CTY High Honors (Score: 625)

WISC-V (2nd Grade)
Extremely High (138 Full Scale, 146 Verbal)

References

Dr. Michael Yetman
Director of Science Curricula and Programs, Proof School

Professor Zvezdelina Stankova
Founder and Director, Berkeley Math Circle, UC Berkeley

ADAM CHEONG

2025: “The Structure & Interpretation of Computer Programs” (CS 61A), UC Berkeley Pre-College Scholar
2025: 4th Place, Biostatistics & Computational Biology, Alameda County Science & Engineering Fair
2025: Honorable Mention, USA Biolympiad (USABO)
2024: “The Mathematics of Competitive Behavior”, Johns Hopkins Center of Talented Youth
2024: Maxima Cum Laude (Silver Recognition), National Latin Examination
2024: Honorable Mention, Zoology, California Science & Engineering Fair
2024: 1st Place & Bill Tobin Award, Life Sciences, Golden Gate STEM Fair
2023: “Logic: Principles of Reasoning”, Johns Hopkins Center of Talented Youth
2023: Honorable Mention, Applied Mechanics & Structures, California Science & Engineering Fair
2023: 1st Place, Engineering: Electrical & Mechanical, Golden Gate STEM Fair
2023: 1st Place, Engineering, Technology & Application of Science, San Mateo STEM Fair
2022: Honorable Mention, Biological Systems, San Mateo STEM Fair

I’ve always loved science. I took biology in freshman year and scored a five on the AP Biology exam. As a sophomore, I am pursuing a “double science” track with AP Chemistry and AP Physics C: Mechanics, along with AP Calculus BC. I have also been a member of the Berkeley Math Circle for 10 years, where I have steadily developed my mathematical reasoning skills.

I am always excited by the challenge of applying what I learn to real-world problem-solving. I am a “founding intern” at Watney Robotics, a startup in San Francisco, where I assemble robot arm controllers for teleoperation, perform data labeling, and conduct ML models training. I aspire to identify and formulate the most important unsolved problems in robotics for our generation, and to create a future in which collective intelligence and resources are mobilized to solve them.

ADAM CHEONG

2025年「The Structure & Interpretation of Computer Programs (CS 61A)」、カリフォルニア大学バークレー校・プレカレッジ・スカラー
2025年 アラメダ郡科学技術フェア、生物統計学・計算生物学部門　第4位
2025年 米国生物学オリンピック (USABO) 努力賞 (Honorable Mention)
2024年「競争行動の数学」、ジョンズ・ホプキンス大学・才能ある青少年センター (CTY)
2024年 全国ラテン語試験 マキシマ・クム・ラウデ (銀賞)
2024年 カリフォルニア州科学技術フェア、動物学部門　努力賞
2024年 ゴールデンゲート STEMフェア、生命科学部門 第1位 & ビル・トービン賞 受賞
2023年「論理学：推論の原理」、ジョンズ・ホプキンス大学・才能ある青少年センター　(CTY)
2023年 カリフォルニア州科学技術フェア、応用力学・構造部門　努力賞
2023年 ゴールデンゲート STEMフェア、電気・機械工学部門 第1位
2023年 サンマテオ STEMフェア、工学・技術・科学応用部門 第1位
2022年 サンマテオ STEMフェア、生物システム部門 努力賞

私は幼い頃から、ずっと科学が大好きです。高校 1年生で生物を履修し、AP Biologyでは5を取得しました。現在は 2年生として、「ダブルサイエンス」として AP ChemistryとAP Physics C: Mechanicsに加え、AP Calculus BCも履修し、より深い学びへと進んでいます。また、 Berkeley Math Circleには10年間在籍し、数学的思考力を着実に培ってきました。

私は常に、学びを現実の問題解決に応用することに強い関心と喜びを感じています。その思いから、サンフランシスコのスタートアップ Watney Robotics にて「創業期インターン」として、ロボットアームや遠隔操作コントローラーの組み立て、データラベリング、機械学習モデルのトレーニングなどに取り組んでいます。

そして私は、私たちの世代におけるロボティクスの最も重要な課題を常に見極め、その解決に向けて英知と資源が結集できる未来を継続的に築いていくことを信念としており、今後もその実現に向けて全力で取り組んでまいります。

3期生
梶田 光
BORN IN 2008
READ MORE
#アート・音楽
#プログラミング
#数学
5期生
大塚蓮
BORN IN 2008
READ MORE
#ロボット
#生物・バイオ
#科学
3期生
SAMAIRA MEHTA
BORN IN 2008
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#コンピュータ
#ビジネス
#教育

Profile Summary

1.	Experience「経験」	Technical Skills「技術力」	Specialty「専門性」
A.	**Watney Robotics** (Silicon Valley Startup)	Robot Arm Controller Assembly, Teleoperation, Data Labeling, ML Models Training, Hydroponics, PCB Soldering	Robotics
B.	**Monte Carlo Simulation** (Open Science)	OpenWorm, Sibernetic, *C. Elegans*, Monte Carlo Method, Decussation, Somatic Twist, Evolutionary Biology	Computational Biology
C.	**Cliffhanger Project** (Mechanical Invention) 	Mountain Climbing Aid, Mechanical Advantage, Type 2 Lever Design, Bouldering, Rock Climbing	Mechanical Engineering

2.	Talented「異能」Outstanding「優れている」
a.	25th International Worm Meeting: Poster Session
b.	2 National Contest Awards:
c.	7 STEM Fair Prizes:
d.	WISC-V: Extremely High (Full Scale IQ: 138)

3.	Wish to Achieve「これから成し遂げたいこと」
a.	18 Math Courses on High School Transcript (Jun 2027)
b.	Research Paper for Regeneron STS (Jun 1~Nov 1 2026)
c.	7 AP STEM Exams: 4 Science, 2 Math, 1 CS (Jun 2026)
d.	UC Berkeley Course (4 Credits): CS 61A (Jun~Aug 2025)